

Data-driven modeling and fault diagnosis for fuel cell vehicles using deep learning

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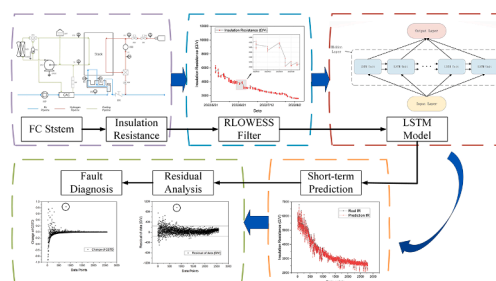
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HIGHLIGHTS

- RLOWESS is used to establish insulation databases of different faults.
- Long short-term memory neural network is applied for insulation prediction.
- The occurrence time points of insulation faults can be effectively identified.

GRAPHICAL ABSTRACT



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ABSTRACT

The reliability and safety of fuel cell vehicle are crucial for the daily operation. Insulation resistance serves as a crucial index of vehicle reliability, especially when fuel cells operate at high voltages. Low insulation resistance can lead to vehicle malfunctions, exposing the operator to the risk of electric shock. In this study, long-term insulation resistance data from thirteen vehicles equipped with three different types of fuel cell systems are analyzed to diagnose possible low insulation resistance issues. For this purpose, a robust locally weighted scatterplot smoothing method is utilized to filter the original data. In this research, an insulation variation model is developed using a data-driven long short-term memory neural network to identify insulation resistance value anomalies caused by deionizer failure. The results indicate that the coefficient of determination of the failure model is 99.78 %. Moreover, current model efficiently identifies insulation faults resulting from reliability issues, such as conductivity issues of cooling pipes and erosion of vehicle wiring harnesses.

1. Introduction

Proton exchange membrane fuel cells (PEMFCs) have been widely applied as a highly promising solution for powering future vehicles due to high efficiency, low noise, and zero emission. Various countries have clearly stated their intention to develop hydrogen energy vigorously.

The Society of Automotive Engineers (SAE) has established durability standards for fuel cell systems in passenger cars, buses, and trucks at 8000 h, 25,000 h, and 30,000 h, respectively [1,2]. Currently, durability and reliability are the primary technical obstacles hindering the large-scale commercialization of fuel cell vehicle (FCV). The main cause of poor reliability is the unavoidable faults that occur during actual

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operation [3]. Studies have shown that data-driven diagnosis methods can accurately and promptly detect faults of fuel cell systems in advance, and can further minimize or eliminate their impact through fault-tolerant control measures [4].

The Insulation resistance (IR) is commonly determined as the resistance of the highest leakage current when the power battery experiences a short circuit with the ground or vehicle chassis [5]. In FCVs, the IR of the entire vehicle is defined as the lower value between the positive and negative insulation resistance of the fuel cell. Because the voltage of fuel cell is up to 350 V SAE requires that the IR of the FCV be at least 500 Ω V⁻¹ to protect vehicle personnel [6]. During FCV operation, the IR value is constantly monitored. If the IR dropped to a certain level, the insulation failure alarm device will be triggered to send an alarm signal and disconnect the high-voltage power. As the time of operating and storage increases, organic and metal ions in the system will gradually release into the coolant [7]. Consequently, this increases the conductivity of the coolant, reducing the IR value of the system and ultimately affecting the insulation of the vehicles. To keep the insulation value within a safe range, the deionizer in the cooling cycle utilizes ion adsorption resin to absorb these ions. However, the resin has a limited adsorption capacity, and degradation of ethylene glycol in the coolant generates organic acids [8], which reduces the lifespan of the resin and results in a gradual decline in insulation performance over time. In summary, operating time, storage time, lifespan of deionizer, and coolant degradation are the primary factors that contribute to decreased insulation over time. In addition, weather and road conditions, as well as fluctuations in system temperature, humidity and vibration [9], may cause certain IR related faults (such as water ingress, wiring harness or auxiliary component issues, among others), leading to local short circuits and a decrease in vehicle insulation. The reliability of fuel cell systems is critical for the commercialization of FCV. Traditional insulation fault diagnosis has the drawbacks of increased human resources, material costs and an inability to meet timeliness and accuracy requirements. Therefore, a new predictive method is necessary to quickly classify fault types and identify faults in advance. This will reduce manual inspection costs while improving the speed and accuracy of diagnostics.

With the advancement of artificial intelligence theory and its practical applications, big data has emerged as a powerful tool for investigating concerns related to fault diagnosis. Currently, in the realm of PEMFC, fault diagnosis methods fall into four categories: model-based methods, data-driven methods, experimental testing methods, and fusion methods [10]. The proportion of research based on data-driven methods has surpassed the other three categories to become the mainstream research direction. Compared to model-based methods, data-driven methods do not require professional knowledge reserves which have the advantage of avoiding complex system modeling. Guo et al. proposed a novel self-learning energy management strategy which has fast and smooth convergence. It can obtain near-optimal results without accurately modeling, without highly relying on experience or prior knowledge [11]. Zeng et al. proposed an iterative learning framework to predict the short-term power demand of vehicle, which can achieve the best prediction performance within the shortest runtime comparing with other learning cores [12]. By using historical sample data, the data-driven method allows the classifier to identify related faults through supervised or unsupervised training. Researchers accomplished fault diagnosis of fuel cell by utilizing various machine learning methods, and have achieved good diagnostic results for various faults within the stack and system. Zhou et al. proposed a new fault diagnosis algorithm BinE-CNN based on binary matrix encoding neural network, which can identify three types of fuel cell faults: drying, flooding, and starvation, with a total accuracy of 0.931 [13]. Lim et al. developed a fault diagnosis model for the cooling subsystem using SVM under three current densities, with an accuracy rate of over 92 % for identifying five types of faults [14]. Zhang et al. proposed a novel method is to classify the different health states, which is composed of simulated annealing genetic algorithm fuzzy c-means clustering and

deep belief network combined with synthetic minority over-sampling technique. This method achieves an accuracy of 99.97 % for training samples and 100 % for testing samples [15]. Nagulapati et al. compared the predictive performance of three data-driven models, GPR, SVM, and ANN, on voltage under dynamic load testing conditions for fault detection and state estimation of PEMFCs [16].

As a type of recurrent neural network (RNN), long short-term memory (LSTM) possesses the exceptional capability of processing time series data, while incorporating a gating mechanism to retain or discard specific features of data. This effectively addresses issues with gradient vanishing and exploding commonly encountered in traditional RNN usage, thereby enhancing the performance of LSTM in the predicting long sequences. In 2018, Rui Ma [17] was the first to utilize LSTM in the context of addressing fuel cell degradation. The proposed Grid-LSTM model achieved a root mean squared error (RMSE) of 0.004 in voltage prediction. By integrating various data processing methods and modifying the network structure, researchers focus on analyzing the LSTM model to achieve high accuracy in predicting voltage. Wang et al. proposed a degradation prediction method based on a LSTM network, the RMSE is 0.5273 for 2000 h test data [18]. Zheng et al. established an LSTM neural network model to achieve polarization curve prediction and performance degradation prediction, with a minimum RMSE of 0.0088 [19]. Liu et al. adopted a LSTM RNN to estimate the remaining life of PEMFC, the prediction accuracy of the novel method is 99.23 %, and the RMSE is 0.003 [20]. Wang et al. established a hybrid model based on semi empirical and data-driven methods to predict the remaining useful life of PEMFC, the RMSE is 0.8144 [21]. Wang et al. built a transient model of PEMFC system and developed a LSTM-MLP deep learning model to predict the output voltage and area specific resistance to monitor the health state of the PEMFC [22]. Implementing LSTM in fault diagnosis provides promptness and is suitable for fault data with temporal characteristics. Consequently, it represents a research avenue for online fault diagnosis. Gu et al. [23] utilized the LSTM model to accurately detect water flooding fault occurrence in the stack by taking hydrogen inlet pressure, air inlet pressure, air inlet flow rate, cooling water inlet and outlet temperature, stack current, and hydrogen recirculation pump current as inputs. Ma et al. [3] used a method based on the fusion of t-Distributed Stochastic Neighbor Embedding and eXtreme Gradient Boosting to identify four types of faults: hydrogen leakage, high hydrogen pressure, low fan speed, and stop purging, with an overall accuracy of 99.31 %. Kim et al. [24] developed a pre-diagnostic system utilizing LSTM and CNN algorithms. The system analyzed time series data, achieving detection rates of 98.52 % for flooding and 95.36 % for drying, within 30 s after their occurrence. The pre-diagnostic system performed optimally within the time frame of 0–60 s.

In the past decade, research on fault diagnosis in fuel cells has primarily focused on the stack interior, with less exploration of insulation faults. Wang et al. [5] proposed a model based insulation fault diagnosis method for the lithium-ion battery pack used in electric vehicles. Hou et al. [9] conducted experimental research on the electrical insulation variation of fuel cell stacks under long-term vibrating conditions through strengthened vibration test. In order to investigate a diagnosis method for insulation faults, LSTM neural networks are used to gather knowledge from previous insulation data and develop models for insulation attenuation. Fault data is input into the model and analyzed with residuals to locate and identify faults, ultimately preventing or minimizing potential damage.

The study is structured in the following order: Section 2 offers an overview of the project background, including the aging and reliability failure mechanisms and manifestations of typical insulation faults in fuel cell systems, as well as raw data preprocessing and filtering methods. Section 3 presents the IR data model based on LSTM neural network. The model is then applied to fault data prediction in Section 4, allowing for successful identification of both fault occurrence time and type. At last, conclusions are given in Section 5.

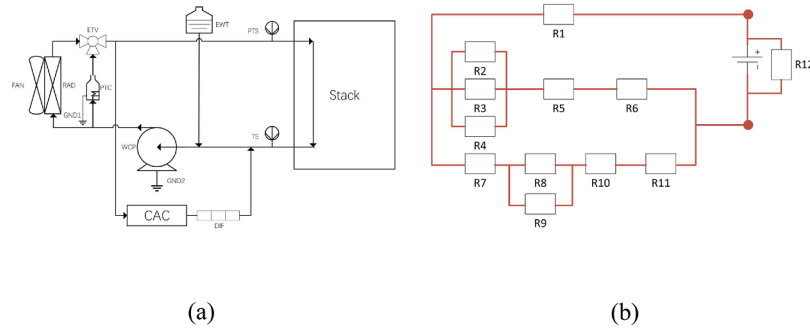


Fig. 1. Structural diagram. (a) Fuel cell cooling system. (b) Equivalent circuit of cooling system in operation.

2. Project background and data preprocessing

2.1. Project background and insulation faults

The data used in this article is derived from the actual driving data of three vehicle fleets on the cloud platform, which equipped with three fuel cell systems from Shanghai Refire Technology Co., Ltd. (REFIRE). Fleet 1 consists of all buses, each equipped with a 50 kW fuel cell system from REFIRE, totaling 8 vehicles. The average operating mileage of the vehicles is 230,000 km, and the average operating time is 10,500 h. Fleet 2 consists of all large buses equipped with a REFIRE 80 kW fuel cell system, totaling 15 vehicles. The average operating mileage of the vehicles is 60,000 km, and the average operating time is 2200 h. Fleet 3 consists of 100 buses, all of which are equipped with a REFIRE 30 kW fuel cell system. The vehicles in Fleet 3 have an average operating mileage of 160,000 km and an average operating time of 7500 h.

The power system of an FCV consists of a fuel cell system and a power cell system in parallel. The structure of the fuel cell system is mainly

composed of stack, hydrogen subsystem, air subsystem, and cooling system. The structural diagram of cooling system is presented in Fig. 1 (a) and is primarily comprised of the following components: water pump (WCP), PTC heater (PTC), radiator (RAD), charge air cooler (CAC), electric temperature valve (ETV), deionizer (DIF), water temperature and pressure sensor (TS, PS), expansion water tank (EWT), and pipeline. During system operation, the pipeline is filled with water. The high-voltage components in the system connected in parallel have a much higher resistance value than the coolant in the cooling pipeline, making the latter negligible. Consequently, IR value of the system equals that of the coolant, and the corresponding calculation formula is [25]:

$$R = \frac{L}{\rho \pi r^2} \quad (1)$$

where L is the length of the cooling pipeline, and r is the radius of the cooling pipeline, ρ is the conductivity of the coolant. Fig. 1(b) depicts the circuit diagram for the cooling system pipeline during operation. The

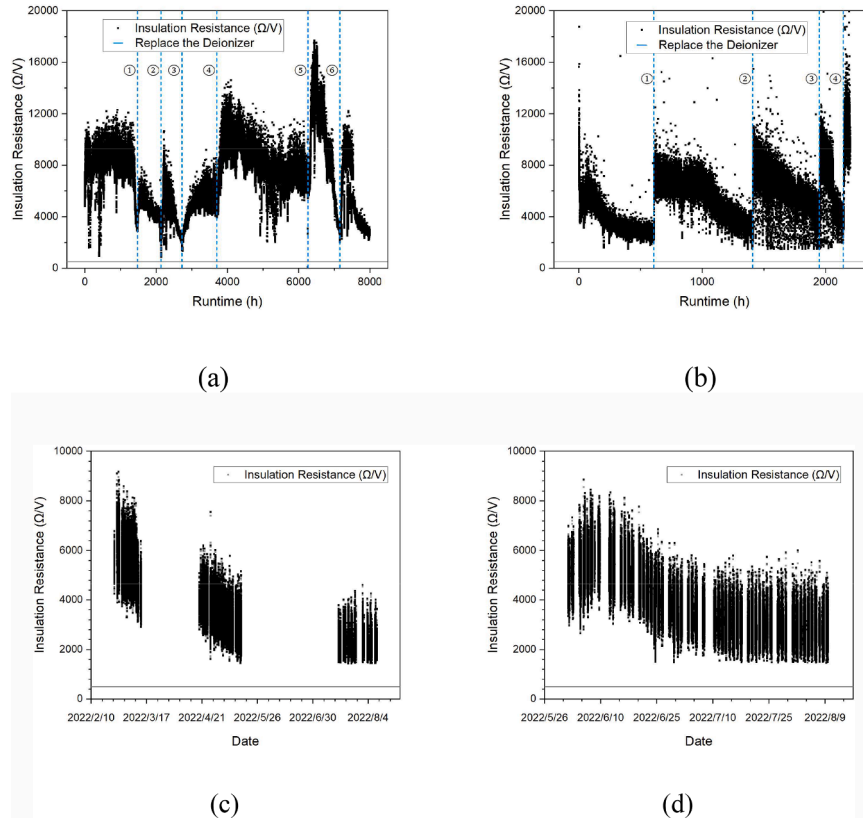


Fig. 2. Diagram of the insulation variation over time: (a) Sample vehicle from Fleet 1. (b) Sample vehicle from Fleet 2. Diagram of IR changes during the life cycle of the deionizer: (c) Vehicle2.1. (d) Vehicle2.3.

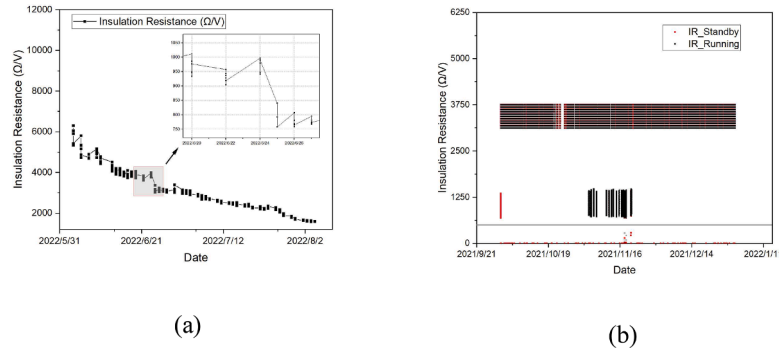


Fig. 3. Diagram of the relationship between IR value and time. (a) For pipeline conducting fault. (b) For wire harness erosion fault.

overall resistance value of the cooling pipeline is calculated by the resistance of each individual pipeline based on their series parallel connection. Eq. (2) provides the formula to obtain the final resistance value.

$$R = \frac{1}{\frac{1}{R1} + \frac{1}{\frac{1}{R2} + \frac{1}{R3} + \frac{1}{R4}} + \frac{1}{R5 + R6} + \frac{1}{\frac{1}{R8} + \frac{1}{R9} + \frac{1}{R7 + R10 + R11}} + \frac{1}{R12}} \quad (2)$$

According to the specifications of the project, in order to improve security, the enterprise standard is generally set much higher than the international standard. However, due to confidentiality issues, the specific values will not be explained in the study.

As stated in the introduction, objective factors causing the low IR values of FCVs while driving can be classified into two categories: deionizer malfunction-induced aging and reliability issues related to IR value failures. To better comprehend the causes of insulation reduction, this study will present the data commencing from the mechanisms and statistics.

Fig. 2(a) and (b) displays the IR data for a sample vehicle from Fleet 1 at 200,000 km and a sample vehicle from Fleet 2 at 50,000 km. The range of the IR data surpasses safety standards (represented by the gray lines in the figure), ensuring vehicle safety. Notably, the figure illustrates some sudden increase nodes, which correspond to deionizer replacement nodes (marked points 1–6 in Fig. 2(a) and 1–4 in Fig. 2(b)). To analyze the characteristics of IR value decline during the life cycle of the deionizer, the IR data of each vehicle was segmented based on the time node of deionizer replacement. Each segment was used as a dataset.

Fig. 2(c) and (d) illustrate the trend of IR value changes for two sample vehicles. Specifically, (c) displays the IR data of Vehicle 1 of Fleet 2 (called Vehicle2.1) during its initial operation, while (d) depicts the IR data between two replacement operations of the deionizer of Vehicle2.3. As illustrated in Fig. 2(c), the IR of a new vehicle initially increases and then decreases over time, owing to long-term storage after leaving the factory. This results lead to the discharge of ions (organic and inorganic) from plastic, aluminum, and rubber components within the cooling system into the coolant [7] consequently increasing the conductivity of the coolant and reducing the IR value of the entire vehicle through leaching, degradation, or corrosion processes. Similarly, the replacement of the deionizer entails a duration during which residual ions are purged from the coolant, leading to a gradual increase in IR over time, as illustrated in Fig. 2(d). In order to analyze the process of IR decrease throughout the lifespan of the deionizer, the corresponding to for each period of IR increase will be excluded, and the remaining data can provide a better representation of the IR changes occurring during the failure process of the deionizer. From Fig. 2, it is evident that IR data exhibits significant fluctuations due to temperature-induced variations in ion activity, and the coolant temperature continuously varies with the operating state of the system, resulting in short-term fluctuations in the insulation data. Additionally, to indicate the effect of placement time on IR data, calendar time is employed as the horizontal axis coordinate.

For the reduction in IR value under standard conditions, it can be classified into two scenarios depending on the operational status of the fuel cell system.

1) Deionizer failure

During the system operation, the ions released by the components diffuse into the coolant, primarily consisting of aluminum ions originating from CAC and RAD in the cooling system. The majority of these ions will be absorbed by the deionizer, with any remaining unabsorbed portions lead to an increase in the conductivity of coolant. The primary component of deionizer is resin, and its adsorption capacity will decrease as the accumulated ion concentration increases. Therefore, over an extended period of time, the IR value of the fuel cell system will gradually decrease. The failure process of the deionizer unfolds over an extended timeframe, exhibiting no discernible daily variations in IR performance. Fault data processing occurs throughout the entire life-cycle of the deionizer, as illustrated in Fig. 2(d). According to product manuals released by companies like Toyota and Hyundai, along with accumulated after-sales experience, the lifespan of the deionizer ranges between 24 and 36 months.

1) Long-term storage

Similar to the failure mechanism of a deionizer, prolonged shutdown can result in a large accumulation of ions in the cooling system. Although the water pump stops working during shutdown, ions can still diffuse into the deionizer due to concentration differences in the coolant, thereby diminishing the lifespan of the deionizer. Consequently, upon restarting after long-term storage, the IR value will be lower than the previous shutdown. As demonstrated in Fig. 2(c), Vehicle2.1 exhibited a decrease of $640 \Omega V^{-1}$ in its IR value when restarted on July 20th after 1580 h of storage, in comparison to the previous shut down.

Occasional faults during the operation of fuel cell can also result in lower IR values, leading to safety hazards and reliability concerns. The following section presents two reasons and their corresponding data manifestations for low insulation values caused by sporadic faults.

1) Cooling pipeline conducting

The pipeline conducting fault in this study is attributed to the breakdown of the water temperature sensor due to high voltage. As depicted in Fig. 1(a), the location of the water temperature sensor is typically at the inlet and outlet of the stack. Due to a broken grounding signal wire of the sensor, the coolant near the sensor flows into the grounding wire, reducing the length of the cooling pipeline (the L in Eq. (1)) in the equivalent IR, and resulting in a decreased IR value, by $800 \Omega V^{-1}$ in the present example. Referring to Fig. 3(a), the IR of Vehicle2.4 decreases suddenly on June 25th. Since the system was operational for several days prior to the fault, insulation degradation is not attributed to

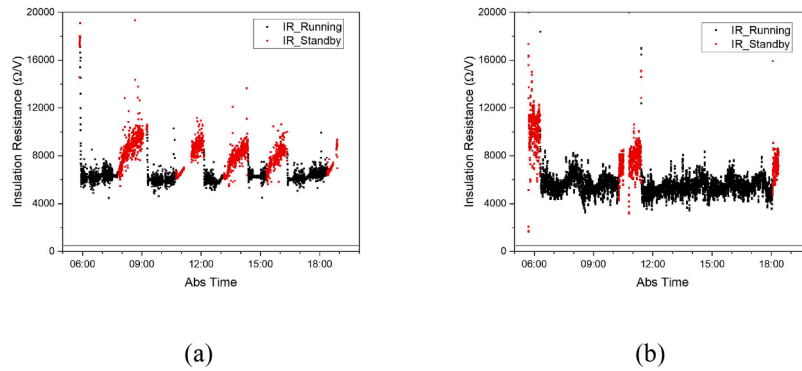


Fig. 4. IR changes during daily operation of the sample vehicle. (a) Fleet 1 (b) Fleet 2.

long-term storage. Upon on-site screening, the fuel cell cooling system water temperature sensor was found to have an IR value of $0 \Omega V^{-1}$.

1) Wire harness erosion fault

The actual road conditions of the vehicles are complex, and continuous vibration may cause loose contact of wiring harnesses in the fuel cell system, which lead to erosion of high-voltage wiring harnesses. During normal driving, the IR range of vehicle3.1 was between 3150 and $3750 \Omega V^{-1}$. One day, due to adhesion between the DCF harness and the FCS negative high-voltage connector, the IR occasionally dropped to a minimum of $800 \Omega V^{-1}$, as shown in Fig. 3(b). This type of failure can cause a sudden disconnection of the high voltage during vehicle operation, resulting in a shutdown of the hybrid mode.

The above introduces several low IR phenomena in fuel cell systems. Next, the raw data will be preprocessed for subsequent analysis.

2.2. Data preprocessing

By dividing each dataset into days, a graph displaying the daily fluctuation of the IR value of the sample vehicle, as shown in Fig. 4. The black dots in the figure represent the IR value of the fuel cell during operation, while the blue ones illustrate the IR value of the fuel cell during standby. It is apparent that Fleet 1 operates for 1–2 h, whereas Fleet 2 operates for 3–7 h per day. In each dataset, a cycle is defined as the period between this powering on and off and the next powering on. The period from powering on to powering off is referred to as the running dataset, while the time from powering off until the next powering on phase is referred to as the standby dataset. While the standby duration does have an impact on the IR value, in the dataset chosen for this study, the standby duration is significantly shorter than the service life of the deionizer. Therefore, the standby dataset is excluded from each cycle, facilitating the analysis of IR changes during vehicle operation. At the same time, in order to better simulate the insulation resistance value of vehicles in normal operation, the data when insulation faults occur is not considered during training. The selected data is

assumed to be from vehicles in normal driving conditions.

From Fig. 2, it is evident that following the fuel cell operation, the high-voltage auxiliary components such as the air compressor and water pump in parallel in the system are sequentially disconnected, causing an increase in the IR value of the entire vehicle. Simultaneously, the coolant temperature gradually decreases to the ambient temperature, leading to a reduction in ion activity and conductivity, resulting in an increase in IR. When powering on again, the connection of high-voltage auxiliary components and the increase in water temperature will lead to a noteworthy decrease in IR value. This decrease fault is evident at the interface between standby and running states, as seen in the figure. Subsequently, there will be significant fluctuations in the data. At the same time, there may be other faults that cause abnormal shutdown in the off state, potentially affecting also affect the IR value. To obtain stable IR data, data is erased after each power-on and for a period of time before shutdown.

From Figs. 2 and 4, it is evident that the original IR data has noise, peaks, and valleys due to potential abnormal power on-off operation and sensor measurement errors. This undesirable effect can result in unwarranted calculations and impede the accuracy of the outcomes. Therefore, it is necessary to filter the original IR data, preserving the trend and remove noise that have no physical significance.

Robust locally weighted scatterplot smoothing (RLOWESS) [26] is a reliable regression technique that can be used for data filtration. To begin with, the weight of each data point in the sample is calculated using the selected kernel function. Then, a weighted least squares approach is utilized to execute first-order polynomial regression for each datapoint in the given window. The weight is adjusted with the residual between the estimated regression value and the original data value. Points with larger residuals receive smaller weights, while data with weights differing from 6 times the average absolute deviation receive a weight of 0. The corrected weights should be used to perform regression again, and the resulting estimated value is then used as the new data point. By employing this method, the information contained in values near the data points can be considered, thereby preserving the trend of the data.

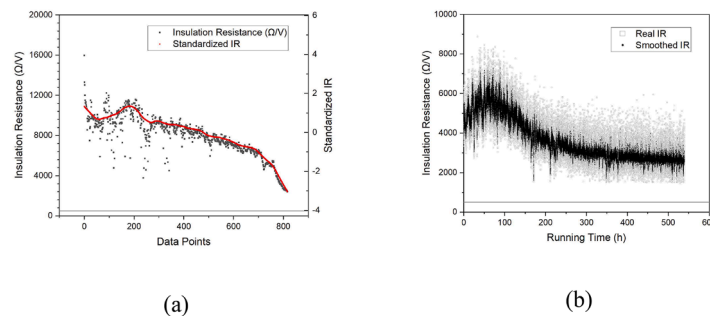


Fig. 5. Comparison of filtered data for sample vehicle. (a) Fleet 1. (b) Fleet 2.

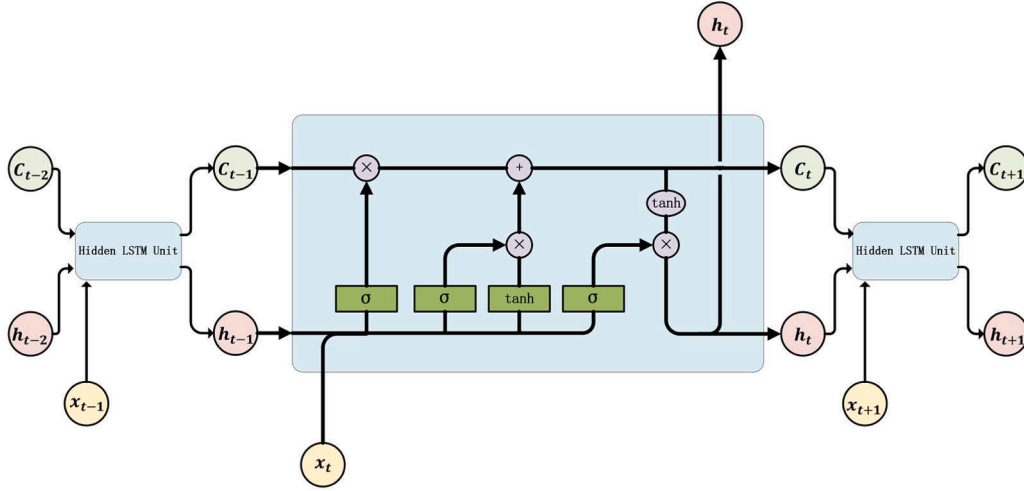


Fig. 6. Architecture of the LSTM memory cell.

The original data utilized in this article was sampled at 1 s intervals. Even after segmentation, the length of long-term operating data ranges from 10^6 to 10^7 . Additionally, excessive data can increase computing costs. To preserve the fundamental features of the data while minimizing its size, it is imperative to utilize the technique of averaging in data processing, which means taking the average of stable operating data based on different time windows. Due to the differences in operating conditions among vehicle fleets, distinct data processing methods will be utilized to analyze the data.

The running time of Fleet 1 is relatively short (Fig. 4(a)). However, each dataset contains a substantial amount of data. Applying the RLOWESS method to the original data would be time-consuming. Therefore, to enhance efficiency, it is suggested to cut and average the data before filtering. Invalid data segments with a running time of less than 30 min are deleted for each operating dataset. After analyzing the duration of single-day insulation stability, we retain the data from the initial 20 min of shutdown for each power-on to shutdown interval and take the average of the insulation data from the initial 10 min as a data point. Reorganize the average data points into a new dataset. Then, apply RLOWESS filtering with a window size equal to 0.1 times the length of the data. Due to the varying IR value ranges in different datasets, the data is scaled through standardization (Eq. (A.2)) to be around 0. This assists with model training and validation since the smoothed data will have a mean of 1 and a standard deviation of 0. Fig. 5 (a) displays one section of the data, both original and filtered. The filtered results had a coefficient of determination (R^2) of 0.82 and a Mean Absolute Percentage Error (MAPE) of 0.06.

The operating time of Fleet 2 is relatively long (Fig. 4(b)), but the data volumes for each are smaller than for Fleet 1. Remove the first 30 min of data after power-on, then consolidate the data from the running cycle and apply smoothing. According to the analysis, the IR value of Fleet 2 remains stable for 90 s with fluctuations that do not exceed $500 \Omega \text{ V}^{-1}$, which resulted in the choice of a 90 s window size. Fig. 5(b) displays the data before and after filtering, with a (R^2) of 0.92 and a MAPE of 0.05 for the filtering results.

The decision coefficient of the filtered data obtained through the aforementioned method is nearly 1, and the MAPE is around 0, indicating that the filtering successfully preserves the trend of the original data. The filtered data will be utilized to construct an LSTM network model.

3. Insulation resistance data model

3.1. Long short-term memory network

The traditional RNN network consists of an input layer, a hidden layer, and an output layer. The neurons in the hidden layer not only accept the input at the current time, but also accept the calculation results of the previous hidden layer. This structure enables RNN to have memory ability, but for long time sequences, this memory ability can lead to gradient vanishing and gradient explosion problems. LSTM is an improved form of RNN, proposed by Hochreiter [27] in 1997.

Compared to the hidden layer of RNN, LSTM has added a cell state, and a single LSTM neuron is shown in Fig. 6, where x_t is the input vector at time t , h_{t-1} and h_t is the output of the previous and current time, C_{t-1} and C_t represents the cell state at the previous and current time. There are three gating units within the neuron, the forget gate (f_t), the input gate (i_t), and the output gate (o_t). Forget gate receives h_{t-1} and x_t , which represents the deletion or retention of information in C_{t-1} . The expression is as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

The input gate receives h_{t-1} , x_t , and C_{t-1} , which represents the processing of information brought by new inputs and combines it with the output information of the forget gate and the previous cell state to obtain the current cell state C_t . The expression is as follows:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (4)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (5)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (6)$$

The output gate receives h_{t-1} , x_t , and C_t , which calculates the output value h_t at the current time. The expression is as follows:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (7)$$

$$h_t = o_t * \tanh(C_t) \quad (8)$$

In Eqs. (3)–(8), W_f, W_i, W_C, W_o represent weight matrix, b_f, b_i, b_C, b_o represents bias vector, $*$ represents bitwise multiplication:

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} * \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 b_1 \\ a_2 b_2 \\ a_3 b_3 \end{bmatrix} \quad (9)$$

$\sigma(\cdot), \tanh(\cdot)$ represent activation function, as follows:

Table 1

Results of different prediction models for fleet 1.

Rank	Model parameter			RMSE of the training set	RMSE of the test set	MAPE of the test set	R^2 of the test set	Training time (s)
	Window size	Number of neurons	Learning rate					
1	10	100	0.01	0.0175	0.0431	0.0242	0.9933	70.7
2	10	100	0.001	0.0177	0.1217	0.0665	0.9465	67.4
3	10	80	0.001	0.0232	0.2570	0.1406	0.7612	69.9
4	10	40	0.001	0.0303	0.2487	0.1390	0.7764	63.4
5	10	80	0.01	0.0312	0.3052	0.1353	0.6633	68.0

Table 2

Results of different prediction models for fleet 1(learning rate = 0.01).

Rank	Model parameter			RMSE of the training set	RMSE of the test set	MAPE of the test set	R^2 of the test set	Training time (s)
	Window size	Number of neurons	Learning rate					
1	2	200	0.01	0.0162	0.1214	0.0713	0.9543	40.6
2	10	100	0.01	0.0175	0.0431	0.0242	0.9933	57.9
3	8	100	0.01	0.0203	0.2024	0.1143	0.8586	51.5
4	4	400	0.01	0.0213	0.5514	0.3361	0.9275	130.1
5	20	40	0.01	0.0218	0.6150	0.3149	0.8935	65.5

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (10)$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (11)$$

According to training requirements, forget gate can affect loop connections by allowing LSTM units to remember or forget the state of the previous step. Meanwhile, if signal data is delivered from the input gate, the current state of the unit can be modified. By filtering the output data, the output gate can determine whether it needs to affect other neurons. Therefore, LSTM can effectively solve the problem of gradient vanishing and gradient explosion in RNN.

In this model, the IR value is used as the input and output parameters, the length of them are 10 and 1. For the IR data produced by various fuel cell systems, the hyperparameters for training LSTM must be selected manually, with a focus on key factors such as the learning rate, number of LSTM hidden layer neurons, data window size, and training batch size. To illustrate, using historical data from Fleet 1, the reduction in IR value during three days of normal operation is less than $120 \Omega V^{-1}$. Therefore, to predict the insulation value for the upcoming operation, a sliding window of three days of data is used. The corresponding average number of operation for three days is ten. Thus, the window size is 10, and the size of sliding step is 1. In the LSTM network structure, 2 layers are initially chosen, and the Adam adaptive optimizer is selected as the optimizer. Based on historical data, we can determine the approximate ranges for certain parameters: the number of neurons should be set to either 20, 40, 60, 80, or 100. By comparing three methods of optimizing learning rates: piecewise decay, exponential decay, and polynomial decay, the learning rate should be set to either 0.1, 0.01, 0.001, or 0.0001. The random seed should be maintained at 1, and the datasets should be split into training set, validation and test sets at an 8:1:1 ratio. After completing the preliminary steps, the model can undergo training by the objective function of accumulate RMSE on the first 24 test points. Afterwards, Table 1 can be consulted for a comparison of the results. After comparison, it was determined that the LSTM network with a learning rate of 0.01 demonstrated the highest level of prediction accuracy for the test dataset. Subsequently, Set the learning rate to 0.01 and continue using the grid search method to find the optimal values for window size and number of neurons, as shown in Table 2. After fixing the remaining hyperparameters, we will compare the training effects of different optimizers in Table 3.

According to the comparison, when the window size is 10, the number of neurons is 100, and the learning rate is 0.01, the model has better training performance. Finally, the aforementioned parameter

Table 3

Training effects of different optimizers.

Method	RMSE of the training set	RMSE of the test set	R^2 of the test set	Training time (s)
Adadelata	0.0232	0.6184	-0.8945	51.8
Adagrad	0.0227	0.4098	0.1678	49.3
Adam	0.0175	0.0667	0.9780	60.2
RMSprop	0.0376	1.2624	-6.8957	53.3

combination was utilized as the model training parameter for Fleet 1.

3.2. Data model

In order to establish a data attenuation model for IR caused by deionizer failure, we assume that the decrease in the IR value is solely related to the decline in deionizer adsorption capacity. The vehicles in Fleet 1 did not exhibit any insulation faults during operation, and their data performance was excellent. Therefore, the IR data from Fleet 1 was selected as the original dataset for training. The IR data of all eight vehicles in Fleet 1 was split to obtain 46 datasets. Referring to the data processing method in Section 2.2, filter and standardize each segment of the dataset. Eventually, we obtained 46 information retrieval datasets, with an average length of 515.

The objective is to identify a dataset that most closely conforms to the IR reduction trend, in order to enhance the predictive capabilities of the trained model for additional data points. To begin with, each data segment is trained, utilizing an 8:1:1 ratio of training, validation, and testing. Using the parameters defined in Section 3.1, a total of 46 LSTM network models were obtained. Subsequently, input the other 45 datasets into the model to calculate RMSE. For every model, calculate the mean and standard deviation of RMSE, then select the smaller one as the final training model. This model will serve as the most representative insulation reduction model. The comparison diagram can be found in Fig. 7(a). Here, Model 7 is selected with a minimum standard deviation of 0.011 and a mean of 0.040. The training set data is depicted in Fig. 5 (a). Fig. 7(b) displays the predicted RMSE of Model 7 for additional data and exhibits effective predictive performance for most datasets.

The data for Model 7 comes from Vehicle1.2, spanning the timeframe from the initial use to the first replacement of the deionizer. The data range, 2000–10,400 ΩV^{-1} , represents the IR range during regular operation in the system. Fig. 7(c) displays the prediction accuracy of Model 7 on the test set with actual errors between 60 and 180 ΩV^{-1} , exerting minimal influence on fault diagnosis. The loss incurred during

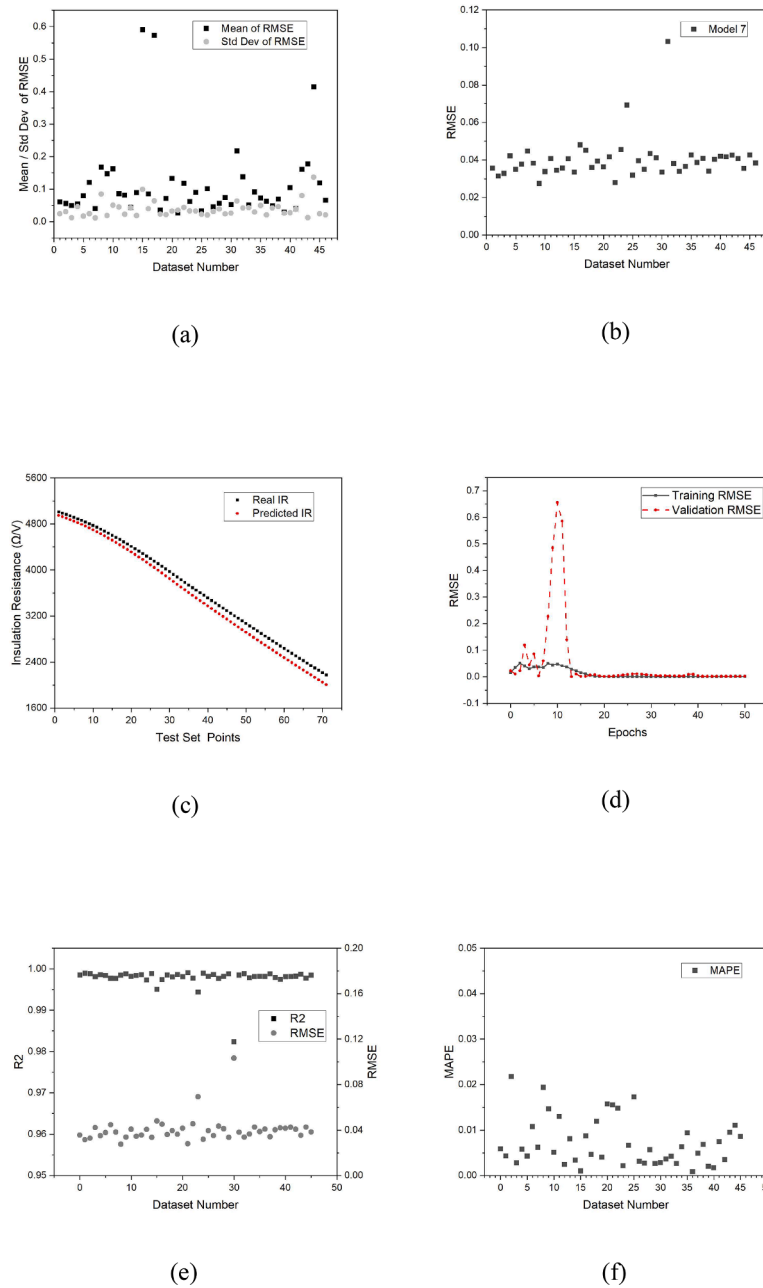


Fig. 7. (a) The predictive performance of each model on other datasets. (b) RMSE of model 7 on other datasets. (c) Prediction of model 7 on the test set. (d) Loss of model 7 at different epochs. (e)-(f): Prediction accuracy and loss of the model on other datasets: (e) RMSE and R^2 , (f) MAPE.

model training is displayed in Fig. 7(d). It is worth noting that the training loss is less than the validation loss, indicating a good fit.

By including all 45 data segments in the prediction, the RMSE for each segment was calculated, as depicted in Fig. 7(e). The average RMSE for other datasets of the model is 0.040, and the lowest RMSE is 0.028. The accuracy of the model is measured by R^2 . The average accuracy of the model is 99.78 %, and the highest accuracy is 99.91 %.

Restoring the predicted data to IR values yielded in an average MAPE of 0.007, with the lowest MAPE being 0.001 (Fig. 7(f)). The difference between the predicted and actual insulation data consistently remains within $180 \Omega V^{-1}$, which is below the range of IR fluctuations in actual data and can be disregarded. This validates the effectiveness of this method in short-term prediction, and the next step is to apply it to specific faults.

4. Method application

Using IR data generated from actual long-term vehicle operation, this study utilizes the low IR value as input to predict any normal or hidden danger states, applying the method outlined in the previous section. By analyzing residual outliers, this approach can predict the time point at which a fault is likely to occur. We will analyze two cases, one related to the long-term storage and cooling pipeline conducting of Fleet 2, and the other concerning wire harness erosion in Fleet 3. For various fuel cell systems, the IR data may differ significantly, but the failure process of the deionizer is similar, allowing for comparison.

4.1. Fault diagnosis for fleet 2

For four vehicles of Fleet 2, extract the IR data between the initial operation and the first deionizer replacement. Apply the filtration

Table 4

Loss of model for different time windows.

Size of time window	RMSE			R^2		
	10 min	30 min	60 min	10 min	30 min	60 min
10 min	0.0309	0.0475	0.0575	0.9774	0.9614	0.9520
30 min	0.0343	0.0487	0.0543	0.9722	0.9595	0.9572
60 min	0.0382	0.0529	0.0622	0.9654	0.9522	0.9438

method detailed in Section 2.2 and compute the average of the filtered data across different time intervals (10, 30, 60 min). Since the selected datasets had similar operational statuses and insulation variation ranges, normalization (Eq. (A.1)) was employed to rescale the data from 0 to 1. Finally, we obtained 12 IR datasets for four vehicles, with average data lengths of 2590, 856, and 396 at varying time intervals.

Use the same approach as in Section 3 to determine that the data

from Vehicle2.3 provides the best fit for the other vehicles. Choose a dataset with varying mean time windows, and split it into a training set and a validation set using a 4:1 ratio. Employ the LSTM network with the identical structure for training, and use the trained model on both itself and the other two data segments. Applying the same methodology as outlined in Section 3, we ascertain that the dataset from Vehicle2.3 offers the most suitable fit for the other vehicles. To accomplish this, a dataset with diverse mean time windows is selected, and it is partitioned into a training set, a validation set and a test set utilizing a 8:1:1 ratio. The LSTM network with an identical structure is employed for training purposes. The trained model is subsequently applied to both the original dataset (Vehicle2.3) and the remaining two datasets. The predicted RMSE and R^2 are presented in Table 4. Objective analysis reveals that a smaller fitted data window leads to a smaller RMSE and a larger R^2 in the results when the training set shares the same window. Adopting an average window value of 10 min after processing the other three

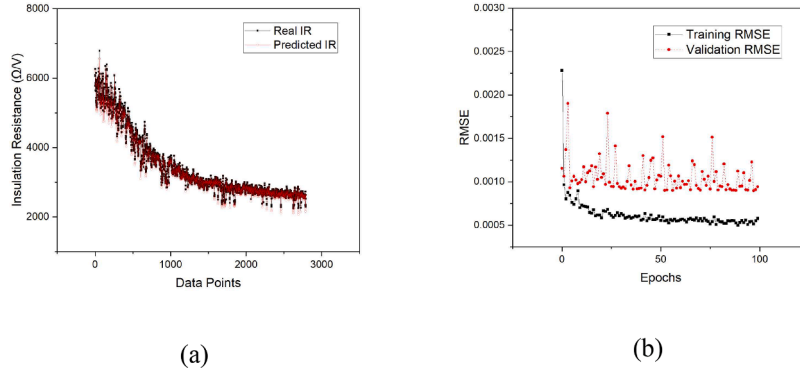


Fig. 8. Training result of model 3. (a) Prediction of model 3 on original dataset. (b) Loss of model 3 at different epochs.

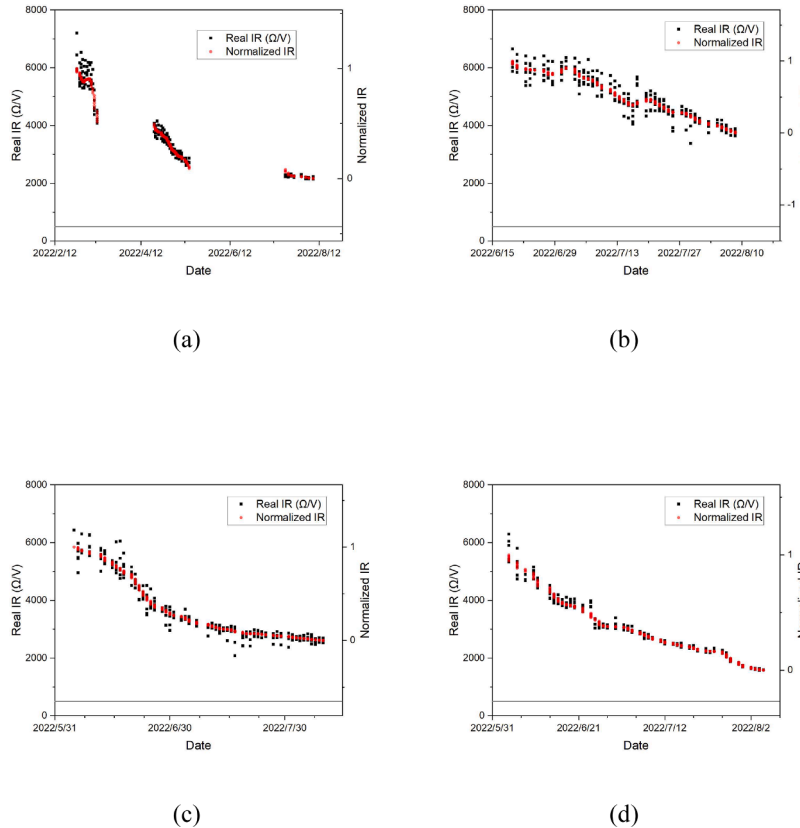


Fig. 9. Comparison of normalized IR data for 4 vehicles of fleet 2.

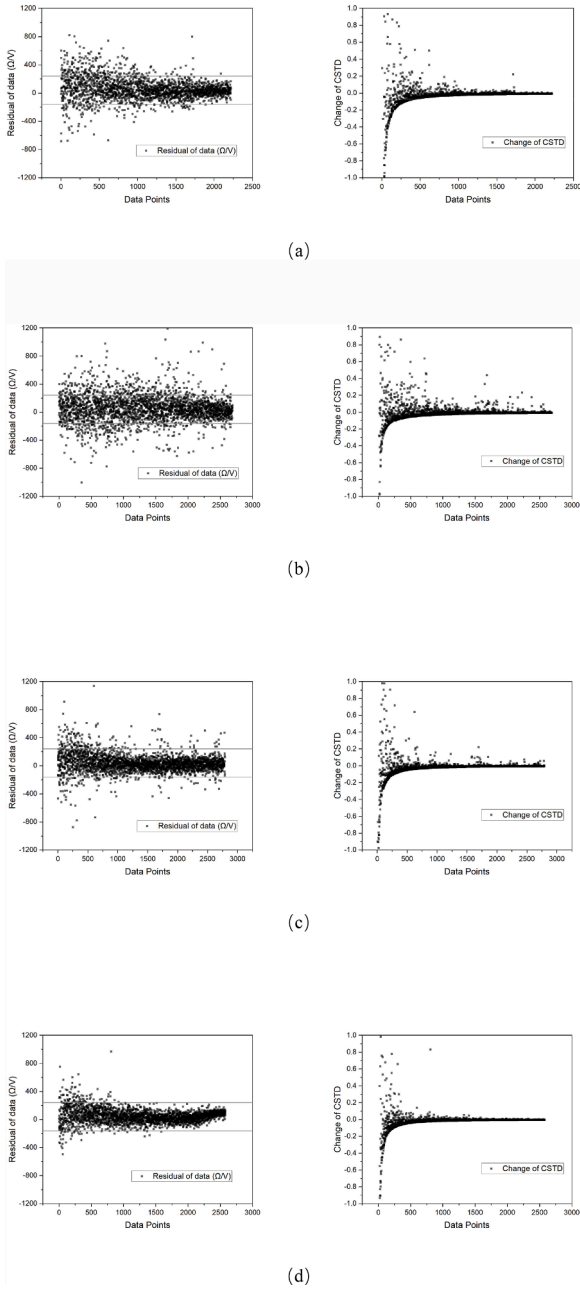


Fig. 10. Comparison of prediction. (a)–(d): Vehicle2.1–2.4. Left: Residual of prediction. Right: Variation of CSTD.

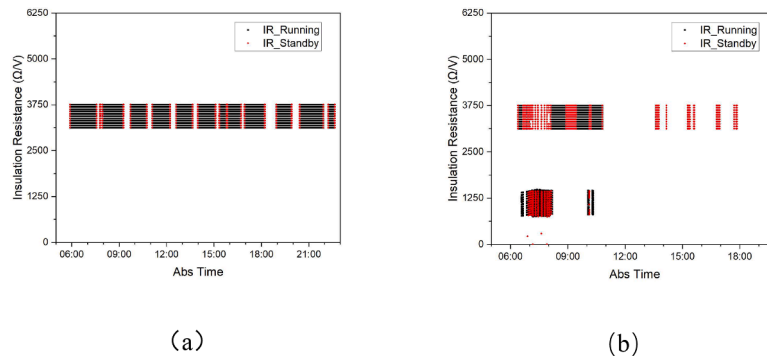


Fig. 11. IR values during daily operation of the sample vehicle from Fleet 3. (a) Normal state (b) Fault state.

datasets with the same approach, the LSTM model was trained with the data from Vehicle2.3. Fig. 8 depict the training effect and error. The data points (the horizontal axis in Fig. 8(a) and subsequent figures) represent the points after data processing, where time information of IR data is saved.

Fig. 9 shows the comparison of data before and after filtering normalization for four datasets, with (a)–(d) respectively corresponding to Vehicles2.1–2.4, respectively. It can be found that there are two storage periods during the operation of Vehicle2.1, corresponding to the phenomenon of IR reduction after long-term storage. During the operation of Vehicle2.2, the IR value of the vehicle exhibited significant fluctuations, serving used as a reference data for large fluctuations in data analysis. The operational data for Vehicle2.3 clearly illustrates the initial failure of the deionizer, with occasional instances of low insulation that are relatively infrequent. During the operation of Vehicle2.4, there was an abnormal point where the IR value suddenly dropped from 4000 to 3200 ΩV^{-1} .

Inputting data for three additional vehicles into the model and adjusting the predicted outcomes to match the actual values the resulting residuals are examined. The left subplots in Fig. 10 display the findings. It is evident that during the initial phase of the failure of the deionizer, the residual fluctuation is considerable. However, as time progresses, the fluctuation diminishes and becomes concentrated around 0. This is because the original data exhibits high volatility in the early stage. Moreover, the daily IR value fluctuation range decreases, corresponding to the gradually failure of deionizer. It is evident that the predicted residuals of the majority of points in the figure lie within the range of -160 – $240 \Omega V^{-1}$. To enhance the visibility of the residual distribution near 0, the standard deviation of the residual data has been calculated. The cumulative standard deviation (CSTD) is calculated by taking the standard deviation of the residuals from all data points preceding the current one, which reflect the deviation of error from the average value. The formula for CSTD is as follows:

$$c_i = \sqrt{\frac{(r_1 - r_i)^2 + \dots + (r_{i-1} - r_i)^2}{i - 1}}, 2 \leq i \leq n \quad (12)$$

Where $(r_1, r_2, \dots, r_n)$ is the residual of the predicted value and the filtered value. When there is a significant variation of CSTD, it can be considered that a fault has occurred at this time. The subplots on the right side of Fig. 10 demonstrate how the variation in CSTD reflects changes in standard deviation. It is evident that the variation value of CSTD decreases gradually as the fluctuation of IR value decreases and eventually approaches 0. Outliers in the data can be distinctly identified by examining the residuals' outliers and CSTD variations.

By comparing Fig. 10(b) and (d) with Fig. 9(b) and (d), it can be found that when the IR value undergoes slight fluctuations (IR performance at the end of the life of the deionizer), the predicted residual fluctuation is relatively small. The variation of CSTD is around 0, with the absolute value not exceeding 0.05. However, when the IR data

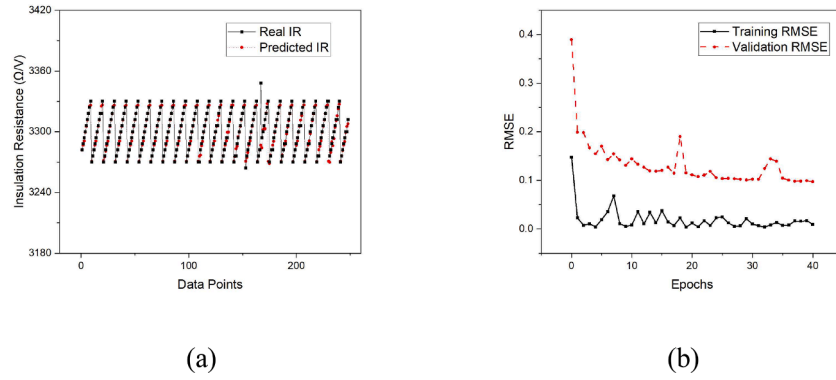


Fig. 12. Training result of model 1. (a) Prediction of model 1 on test set. (b) Loss of model 1.

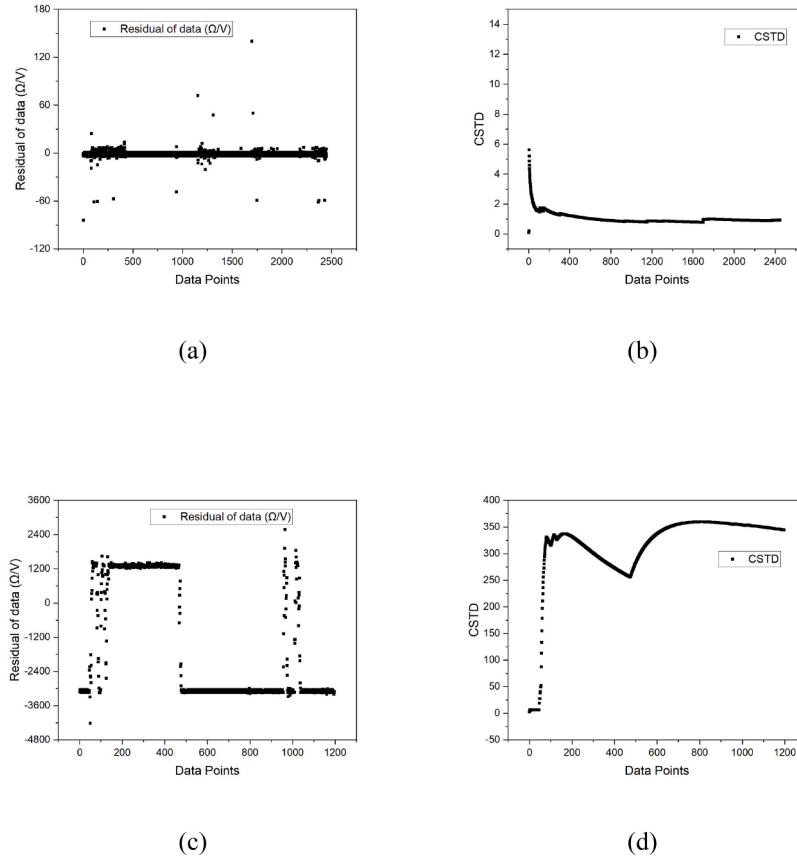


Fig. 13. (a) Predicted residual of normal data. (b) CSTD of normal data. (c) Predicted residual of fault data. (d) CSTD of fault data.

exhibits significant fluctuations, the variation of CSTD is large, and the absolute value often exceeds 0.2.

From the comparison results shown in Fig. 9, it is evident that the predicted results reflect the causes of IR reduction. Specifically, at the 1,713th data point, the CSTD fluctuation of Vehicle2.1 is relatively small but still greater than 0.2, indicating the presence of abnormal data. Consistent with the actual data, the IR reduction is caused by prolonged storage. The residual fluctuation of Vehicle2.2 is significantly greater than that of typical vehicles in the same fleet, resulting in a large fluctuation of IR in the actual data. However, the reason for this phenomenon is still unknown. The average residual of Vehicle2.4 is less than 800 ΩV^{-1} , but an abnormal point with a residual of 1000 ΩV^{-1} appears at the 807th data point. At the same time, the variation of CSTD exceeds 0.8, which is significantly higher than the normal level. The pipeline conducting fault is believed to be the cause of the corresponding actual

data.

4.2. Fault diagnosis for fleet 3

In contrast to the first two fleets, Fleet 3 exhibits a distinctive range of IR values due to the difference in insulation detection modules. To obtain the anticipated results from the model, diverse methods of data processing are necessary. Fig. 11(a) illustrates the normal running IR value, which fluctuates between 3125 and 3750 ΩV^{-1} . In the event of a fault, the IR value will decrease to 625–1625 ΩV^{-1} . Restarting after shutdown will restore the IR, but low IR will reoccur subsequently. Refer to Fig. 11(b) for IR data on the day of the fault.

Based on this phenomenon, the operational data of Vehicle 3.1 during the normal driving phase on a specific day is initially pre-processed. To enhance fault diagnosis, a 10 s time window is employed

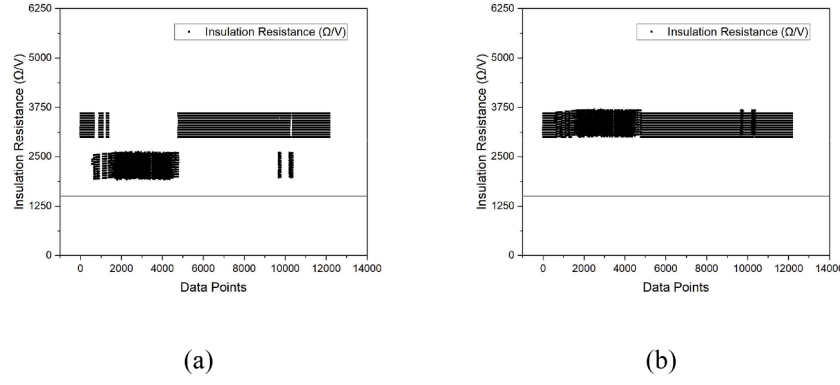


Fig. 14. Fault data above detection threshold: (a) Fault example 1. (b) Fault example 2.

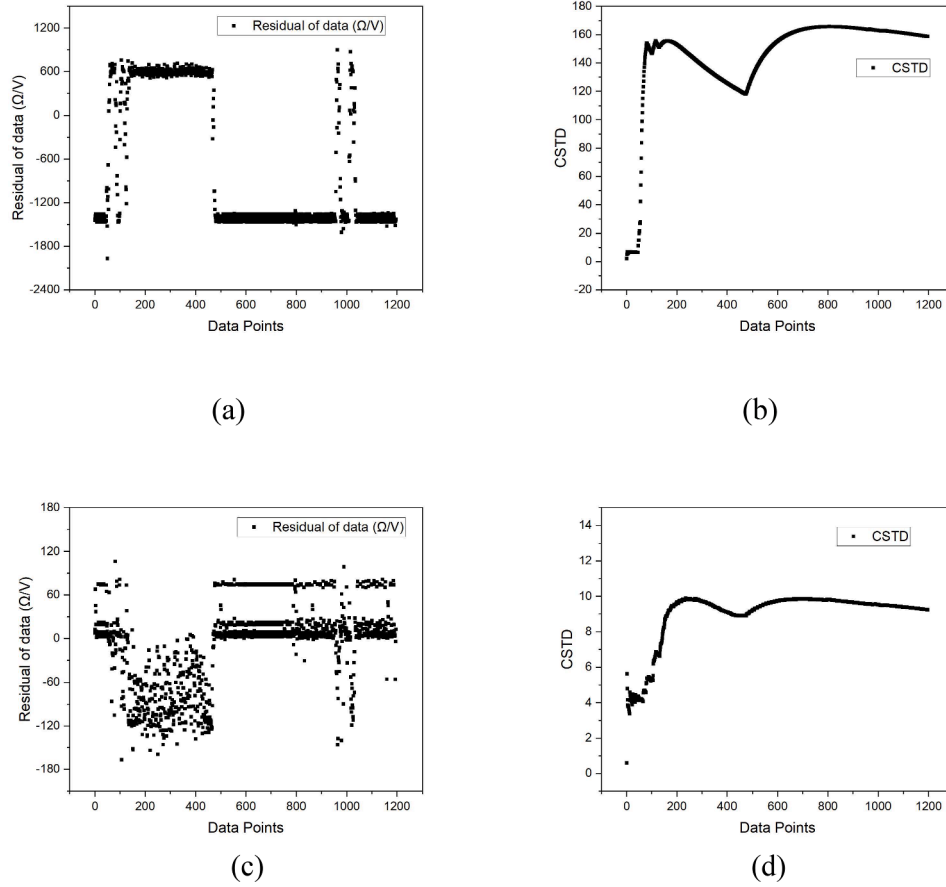


Fig. 15. (a) Predicted residual of fault example 1. (b) CSTD of fault example 2. (c) Predicted residual of fault example 1. (d) CSTD of fault example 2.

to average the original data considering the similar trend of the IR value feedback from the insulation module of Fleet 3 based on historical experience. The data sampled each time is not included in the next averaging, and the length of the final normal operation data set is 2469. Divide the average data into a 8:1:1 ratio between a training set, a validation set, and a test set, then employ the same structure of an LSTM network for training. The resulting RMSE is less than 0.02, indicating that the model can capably fit the vehicle data during normal operation, and this model is denoted as Model 1. The training effectiveness and loss of the model are presented in Fig. 12(a) and (b).

After obtaining Model 1, input the daily data for both normal driving and faulty driving into the model to make short-term predictions. After the predicted results are restored to the true values, the predicted residuals are analyzed, and the statistical results are shown in Fig. 13.

From Fig. 13(a)–(d), it is evident that the predicted residuals during regular driving vary from -90 to $160 \Omega V^{-1}$, mainly concentrated in -20 – $20 \Omega V^{-1}$. The CSTD decreases gradually and eventually stabilizes around 1. When a fault transpires, the predicted residual falls between -3750 and $1850 \Omega V^{-1}$, and the CSTD surpasses 250. By comparing the CSTD, it is apparent that a harness ablation fault transpired on that day. The drop in IR value from the harness ablation fault always remains below $1500 \Omega V^{-1}$, which is lower than the alarm threshold of the detection module. Hence, it can be identified and triggered by the fault detection module within the system. Subsequently, the scenario where the low insulation value of the fault surpasses the alarm threshold will be analyzed.

If the IR value caused by the fault decreases and falls below the alarm threshold of the detection module by $1500 \Omega V^{-1}$, this method can still

identify the fault occurrence. By maintaining the normal driving data and increasing the IR value of the entire faulty data segment, the IR value of the faulty data is kept above $1500 \Omega V^{-1}$, which is referred to as Fault Example 1. Furthermore, maintaining an IR value of $3150\text{--}3750 \Omega V^{-1}$ for the fault data is referred to as Fault Example 2. Fig. 14 displays the new fault data.

Fig. 15 displays the predicted residuals and CSTD for two fault scenarios. When compared to the normal state results shown in Fig. 13(a) and (b), Fault Example 1 exhibits a relatively large residual fluctuation range from -1550 to $950 \Omega V^{-1}$, with a CSTD exceeding 110. Fault Example 2, on the other hand, shows a residual fluctuation range similar to normal driving, with a range of $-200\text{--}120 \Omega V^{-1}$. However, the residual values predicted for IR values during the period without any fault occurring on the day of the fault are primarily concentrated between $0\text{--}30$ and $60\text{--}90 \Omega V^{-1}$. The fault occurrence will decrease the residual to $-140\text{--}0 \Omega V^{-1}$. Concurrently, the CSTD of Fault Example 2 exceeds 8 and reaches a stable rate of around 10, which differs significantly from the anticipated residual and CSTD during regular driving conditions, and it is identifiable.

Therefore, it can be concluded that the model accurately identifies fault data below $3125 \Omega V^{-1}$, indicating its ability to identify long-term reduction of IR values caused by other possible faults. Additionally, the model can detect faults if the insulation fluctuation range is the same as normal data. The practical significance of the research findings lies in the fact that the alarm threshold for vehicle insulation typically stands at $1900 \Omega V^{-1}$. If a particular fault causes a drop in the IR to a value above the alarm threshold, the vehicle will have a low IR but no alarm, posing a safety risk. By analyzing daily insulation data and promptly identifying faults, this method can contribute to hazard prevention.

5. Conclusion

This article presents an overview of the mechanism and data representation of three occurrences of low insulation value in fuel cell vehicles: deionizer failure, wire harness erosion, and cooling pipeline conducting. A data-driven Proton exchange membrane fuel cell insulation fault diagnosis method based on LSTM neural network is proposed. This approach creates a data model for insulation reduction stemming from deionizer failure using raw data filtered by the RLOWESS method. The conclusions are as follows:

1. The model provides dependable predictions for the reduction in insulation due to the decreased lifespan of the deionizer, achieving a maximum accuracy of 99.91 % and a minimum MAPE of 0.001. The residual of the predicted data is under $200 \Omega V^{-1}$, making it negligible in practical applications.

Appendix

Explain the formula and loss function utilized in this study:

Prior to feeding data into the neural network training, it is imperative to normalize and standardize to mitigate the influence of data dimensionality on the outcomes. For input data $x = (x_1, x_2, \dots, x_n)$, the formulas for normalization and standardization are as follows:

Normalization:

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (A1)$$

Standardization:

$$x'_i = \frac{x_i - \bar{x}}{x_{std}} \quad (A2)$$

where $x_{\min}$ is the minimum value of x , $x_{\max}$ is the maximum value of x , $\bar{x}$ is the average value of each component of x , x_{std} is the standard deviation of x .

To evaluate the predictive performance of the model, we have selected the following three indicators:

RMSE

2. Applying this method to insulation abnormal data can determine the time nodes of long-term storage and pipeline conducting faults. A comparison of the manifestation of accumulated data on insulation faults can aid in the manual identification of fault types.
3. After training on historical data, the model predicts new daily data in under 1 s, offering a potential means for real-time fault detection.

The data used in this article focuses on the operating state of the fuel cell system, ignoring the insulation data in standby and shutdown states, as well as insulation changes caused by faults other than those mentioned in the article. Future work will add variables such as storage time and temperature changes that may cause insulation values to change. Simultaneously, if there exists a substantial volume of historical vehicle operating data, the application of the RLOWESS approach may demand significant time for initial model training, posing challenges for the analysis of multiple fleets in batches. In subsequent efforts, we plan to enhance the multivariate prediction model and research methods to minimize training time. This objective remains the focus of our forthcoming endeavors.

CRediT authorship contribution statement

Yangeng Chen: Software, Validation, Visualization, Writing – original draft. **Jingjing Zhang:** Conceptualization, Methodology, Supervision. **Shuang Zhai:** Conceptualization, Data curation, Methodology, Resources, Supervision, Validation, Writing – review & editing. **Zhe Hu:** Data curation, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (A3)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (\bar{y}_i - y_i)^2} \quad (A4)$$

After restoring the predicted data to actual values, use the Mean Absolute Percentage Error (MAPE) to assess the disparity between the predicted and actual values.

$$MAPE = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|}{|y_i|} \quad (A5)$$

References

- [1] Fuel cell technologies office multi-year research, development, and demonstration plan - section 3.4 fuel cells. 2017; www.energy.gov/sites/prod/files/2017/05/f34/fcto_myrrdd_fuel_cells.pdf (accessed July 21, 2023).
- [2] DOE. Hydrogen and fuel cells program record 19006: hydrogen class 8 long haul truck targets. 2019; https://www.hydrogen.energy.gov/pdfs/19006_hydrogen_class8_long_haul_truck_targets.pdf (accessed July 21, 2023).
- [3] Ma R, Dang H, Xie R, Xu L, Zhao D. Online fault diagnosis for open-cathode PEMFC systems based on output voltage measurements and data-driven method. *IEEE Trans Transp Electrification* 2022;8:2050–61. <https://doi.org/10.1109/TTE.2021.3114194>.
- [4] Dijoux E, Steiner NY, Benne M, Péra M-C, Pérez BG. A review of fault tolerant control strategies applied to proton exchange membrane fuel cell systems. *J Power Sources* 2017;359:119–33. <https://doi.org/10.1016/j.jpowsour.2017.05.058>.
- [5] Wang Y, Tian J, Chen Z, Liu X. Model based insulation fault diagnosis for lithium-ion battery pack in electric vehicles. *Measurement* 2019;131:443–51. <https://doi.org/10.1016/j.measurement.2018.09.007>.
- [6] Recommended practice for general fuel cell vehicle safety. J2578.201408 2014; https://www.sae.org/standards/content/j2578_201408/ (accessed July 21, 2023).
- [7] Mueller S.A., Kim B.R., Anderson J.E., Kumar M., Huang C. Leaching of ions from fuel cell vehicle cooling system and their removal to maintain low conductivity. 2003; p. 2003-01–0802. <https://doi.org/10.4271/2003-01-0802>.
- [8] Kamo K., Miyashita M., Study on thermal degradation characteristics of ion exchange resins of fuel cell vehicles. 2019; p. 2019-01–2370. <https://doi.org/10.4271/2019-01-2370>.
- [9] Hou Y, Zhou W, Shen C. Experimental investigation of gas-tightness and electrical insulation of fuel cell stack under strengthened road vibrating conditions. *Int J Hydrog Energy* 2011;36:13763–8. <https://doi.org/10.1016/j.ijhydene.2011.07.092>.
- [10] Rui Ma, Hanbin Dang, Yuqi Zhang, Zhe Huo, Yuren Li. A Review on failure mechanism analysis and diagnosis for proton exchange membrane fuel cell systems. *Proc CSEE* 2022;1–21. <https://doi.org/10.13334/j.0258-8013.pcsee.222031>.
- [11] Guo L, Li Z, Outbib R, Gao F. Function approximation reinforcement learning of energy management with the fuzzy REINFORCE for fuel cell hybrid electric vehicles. *Energy AI* 2023;13:100246. <https://doi.org/10.1016/j.egyai.2023.100246>.
- [12] Zeng T, Zhang C, Hao D, Cao D, Chen J, Chen J, et al. Data-driven approach for short-term power demand prediction of fuel cell hybrid vehicles. *Energy* 2020;208:118319. <https://doi.org/10.1016/j.energy.2020.118319>.
- [13] Zhou S, Lu Y, Bao D, Wang K, Shan J, Hou Z. Real-time data-driven fault diagnosis of proton exchange membrane fuel cell system based on binary encoding convolutional neural network. *Int J Hydrog Energy* 2022;47:10976–89. <https://doi.org/10.1016/j.ijhydene.2022.01.145>.
- [14] Lim IS, Park JY, Choi EJ, Kim MS. Efficient fault diagnosis method of PEMFC thermal management system for various current densities. *Int J Hydrog Energy* 2021;46:2543–54. <https://doi.org/10.1016/j.ijhydene.2020.10.085>.
- [15] Zhang X, Zhou J, Chen W. Data-driven fault diagnosis for PEMFC systems of hybrid tram based on deep learning. *Int J Hydrog Energy* 2020;45:13483–95. <https://doi.org/10.1016/j.ijhydene.2020.03.035>.
- [16] Nagulapati VM, Kumar SS, Annadurai V, Lim H. Machine learning based fault detection and state of health estimation of proton exchange membrane fuel cells. *Energy AI* 2023;12:100237. <https://doi.org/10.1016/j.egyai.2023.100237>.
- [17] Ma R, Yang T, Breaz E, Li Z, Briois P, Gao F. Data-driven proton exchange membrane fuel cell degradation predication through deep learning method. *Appl Energy* 2018;231:102–15. <https://doi.org/10.1016/j.apenergy.2018.09.111>.
- [18] Wang Y, Wang K, Wang B, Yin Y, Zhao H, Han L, et al. A data-driven approach to lifespan prediction for vehicle fuel cell systems. *IEEE Trans Transp Electrification* 2023;1. <https://doi.org/10.1109/TTE.2023.3237219>.
- [19] Zheng L, Hou Y, Zhang T, Pan X. Performance prediction of fuel cells using long short-term memory recurrent neural network. *Int J Energy Res* 2021;45:9141–61. <https://doi.org/10.1002/er.6443>.
- [20] Liu J, Li Q, Chen W, Yan Y, Qiu Y, Cao T. Remaining useful life prediction of PEMFC based on long short-term memory recurrent neural networks. *Int J Hydrog Energy* 2019;44:5470–80. <https://doi.org/10.1016/j.ijhydene.2018.10.042>.
- [21] Wang Y, Wu K, Zhao H, Li J, Sheng X, Yin Y, et al. Degradation prediction of proton exchange membrane fuel cell stack using semi-empirical and data-driven methods. *Energy AI* 2023;11:100205. <https://doi.org/10.1016/j.egyai.2022.100205>.
- [22] Wang B, Yang Z, Ji M, Shan J, Ni M, Hou Z, et al. Long short-term memory deep learning model for predicting the dynamic performance of automotive PEMFC system. *Energy AI* 2023;14:100278. <https://doi.org/10.1016/j.egyai.2023.100278>.
- [23] Gu X, Hou Z, Cai J. Data-based flooding fault diagnosis of proton exchange membrane fuel cell systems using LSTM networks. *Energy AI* 2021;4:100056. <https://doi.org/10.1016/j.egyai.2021.100056>.
- [24] Kim K, Kim J, Choi H, Kwon O, Jang Y, Ryu S, et al. Pre-diagnosis of flooding and drying in proton exchange membrane fuel cells by bagging ensemble deep learning models using long short-term memory and convolutional neural networks. *Energy* 2023;266:126441. <https://doi.org/10.1016/j.energy.2022.126441>.
- [25] Chen X, Wang X, Sun W, Jiang C, Xie J, Wu Y, et al. Integrated interdigital electrode and thermal resistance micro-sensors for electric vehicle battery coolant conductivity high-precision measurement. *J Energy Storage* 2023;58:106402. <https://doi.org/10.1016/j.est.2022.106402>.
- [26] Cleveland WS. Robust locally weighted regression and smoothing scatterplots. *J Am Stat Assoc* 1979;74:829–36. <https://doi.org/10.1080/01621459.1979.10481038>.
- [27] Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput* 1997;9:1735–80. <https://doi.org/10.1162/neco.1997.9.8.1735>.